

Day 6 — Why It Works

Module: Nikhilam Subtraction · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will explain *why* the Nikhilam rule produces the correct answer, using the identity $10^n - A = (10^n - 1) - A + 1$, and will draw a bar-model that shows the same idea visually.

Materials

- Whiteboard
- Worksheet: “explain to a friend” prompt + space
- Coloured pencils for the bar model
- The 16-sūtra list (Microsoft Word - sutras1.pdf) visible

Lesson flow

Warm-up (5 min)

- Sūtra chant.
- Quick fire: 4 complements (mix of 2-digit and 3-digit).
- Return graded formatives. Acknowledge the common errors.

Core (20 min) — The mathematical reason

1. **The core identity (write boxed on the board):**

$$10^n - A = (10^n - 1) - A + 1$$

2. **Why this identity?** “ 10^n is 1 followed by n zeros — e.g. $1000 = 10^3$. $10^n - 1$ is the same number minus one — e.g. 999. That’s just ‘all nines.’ So the identity says: subtracting A from a round base equals (subtracting A from all-nines) plus 1.”
3. **Why does that help?** “Subtracting any digit from 9 is trivial — there’s no borrow. We always have $9 \geq \text{digit}$. So $(10^n - 1) - A$ is a clean digit-by-digit subtraction with no borrow. Then we add 1 at the end.”
4. **Where does the “+1” go?** “Into the last digit. So instead of doing ‘from 9’ on the last digit and then ‘+1,’ we just do ‘from 10’ on the last digit. Same thing. That’s why the sūtra says ‘all from nine and **the last from ten.**’”
5. **Walk through a concrete example with full algebra:**

$$\begin{aligned} &1000 - 567 \\ &= (1000 - 1) - 567 + 1 && \leftarrow \text{identity} \\ &= 999 - 567 + 1 \\ &= (9-5)(9-6)(9-7) + 1 && \leftarrow \text{digit-wise from 9 — no borrow!} \end{aligned}$$

$$\begin{aligned}
 &= 4 \ 3 \ 2 + 1 \\
 &= 432 + 1 \\
 &= 433. \quad \checkmark
 \end{aligned}$$

Now compare with the sūtra direct version:

$$\begin{aligned}
 &1000 - 567 \\
 &= (9-5)(9-6)(10-7) \quad \leftarrow \text{Nikhilam, last from 10} \\
 &= 4 \ 3 \ 3 \\
 &= 433. \quad \checkmark
 \end{aligned}$$

“Same answer. The sūtra is just the identity, made into a mnemonic.”

6. Bar model.

Draw a long horizontal bar labelled 1000. Inside, mark off a left segment labelled 567 and a right segment labelled ? (the unknown). *“What we want is the right segment.”*

Now ask: *“What’s the length of the bar minus 1?”* — Answer: 999.

Mark a tiny +1 sliver at the right end of the bar. Now $999 = 567 + (\text{right segment} - 1)$, so right segment $= (999 - 567) + 1 = 432 + 1 = 433$.

“That’s the picture-version of the algebra. Same thing.”

Activity (15 min) — Explain to a friend

- Worksheet prompt:
Imagine your Grade 4 cousin asks: *“Why does Nikhilam work? Isn’t it just magic?”*
In 4–6 sentences, explain why it’s NOT magic. Use the words **nine**, **ten**, **plus one**, and **place value**.
- 10 min individual writing.
- 5 min pair-share: each partner reads their paragraph; partner says one thing that’s clear and one that’s unclear.

Wrap-up (5 min)

- Read one strong paragraph aloud (with permission).
- Honest historical note: *“Tirthaji’s book (1965) gives this rule. The rule itself is genuinely correct. Tirthaji’s claim that it comes from a Vedic appendix has not been verified by other Sanskritists. We use the rule because it works. The history is younger than the name suggests.”*
- Preview Day 7: *“Tomorrow — a surprise. The same rule is built into how computers subtract. It’s called ‘two’s complement.’”*

Student-facing content

Why Nikhilam works

The key identity:

$$10^n - A = (10^n - 1) - A + 1$$

- $10^n - 1$ is “all nines” (e.g. 9999).
- Subtracting A from “all nines” is trivial digit-by-digit — there’s no borrow because $9 \geq$ any digit.
- The “+ 1” at the end gets absorbed into the last digit, turning “9 – last_digit” into “10 – last_digit.”

That’s the whole proof.

Tirthaji’s *Vedic Mathematics* (1965) packages this proof into six Sanskrit words: *Nikhilam Navataścaramaṁ Daśataḥ* — “all from nine and the last from ten.” The math is real. The Sanskrit is from a 20th-century synthesis, not a Vedic text.

It is not magic. It is good notation.

Homework

- Find ONE moment in your daily life where you naturally “round to the nearest 100 / 1000” without thinking. Write 2 sentences describing it. (E.g. tipping at a restaurant. Counting change. Comparing prices.)
- Practise: complement of 4321 to 10 000. Show the algebraic identity for this one (the same way the lesson did $1000 - 567$).

Differentiation

- **Tier up:** Write the explanation for a Grade 11 student. Use formal notation. Show the identity holds for any base b : $b^n - A = (b^n - 1) - A + 1$. Preview Day 7’s binary case.
- **Tier down:** Skip the written explanation. Verbalise to your partner. Use the bar-model picture as scaffold.

Teacher notes

- The most common student error in the paragraph: confusing “the last digit takes from 10” with “you add 1 at the end.” These are the SAME operation, just expressed differently — $10 - x = (9 - x) + 1$. Some students will write both; that’s fine.
- Some students prefer the visual bar-model; some prefer the algebra. Both are valid. Honour the choice and ask them to draw the OTHER one as homework.
- Honest provenance is essential here. Don’t oversell. Some students (or parents) believe Nikhilam is “ancient Vedic mathematics.” Clarify gently; preview the Senior-band treatment of the historicity question.

Citation(s) used in this lesson

- Tirthaji, *Vedic Mathematics* (1965), sūtra Nikhilam (#2) — primary source.
- *Microsoft Word - sutras1.pdf* — wall reference for the full 16 sūtra list.